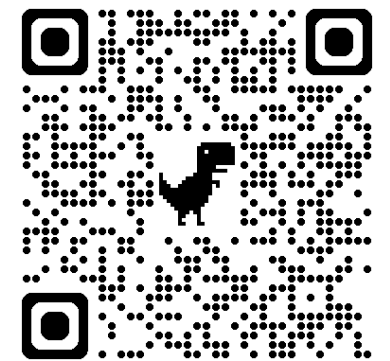




DVU group **Data, Vibration, and Uncertainty Group**

DATA-CENTRIC ENGINEERING



Schedule of today:

Probabilistic and Physics-Informed Machine Learning

09:00–10:00: L1 – Frontal Lecture (intro): Probabilistic Machine Learning (PML)

Coffee break: 10:00–10:30

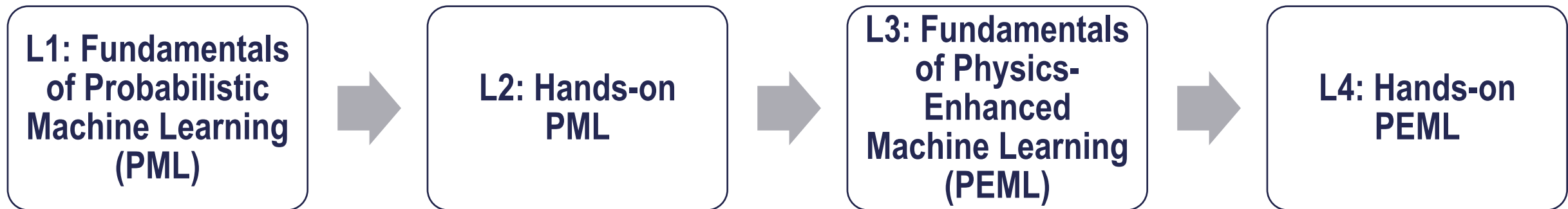
10:30–12:00: L2 – Mostly Hands-on PML (Jupyter notebooks)

Lunch: 12:00–14:00

14:00–15:00: L3 – Frontal lecture (intro): Fundamentals of Physics-Enhanced Machine Learning (PEML)

Coffee break: 15:00–15:30

15:30 - 17:00: L4: Mostly Hands-on PEML (Jupyter notebooks)



Lecturer: Alice Cicirello (ac685@cam.ac.uk)

Fundamentals of Probabilistic Machine Learning

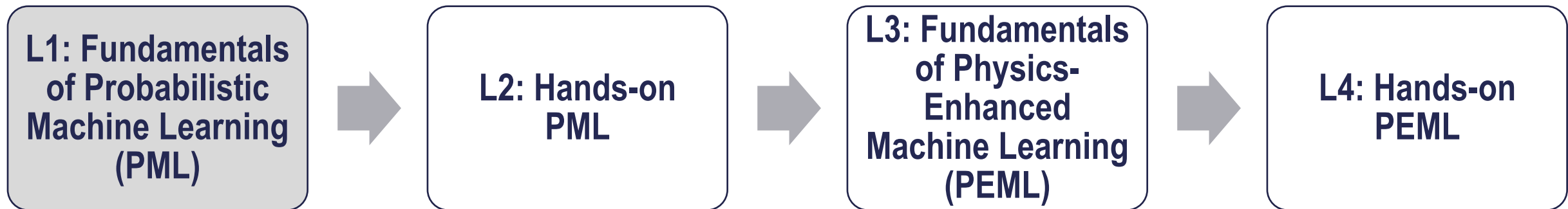
Lecturer: Alice Cicirello (ac685@cam.ac.uk)

September 2026

Learning objectives of this lecture

L1: Fundamentals of Probabilistic Machine Learning

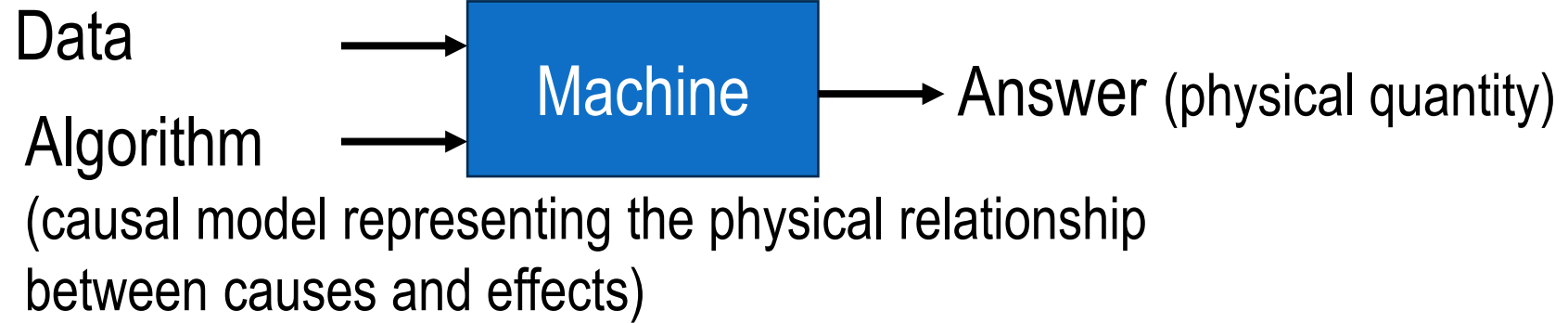
- Machine Learning vs Probabilistic Machine Learning
- Sources of uncertainties – with a practical example (notebook)
- Fundamentals of probabilistic analysis (all in a notebook)



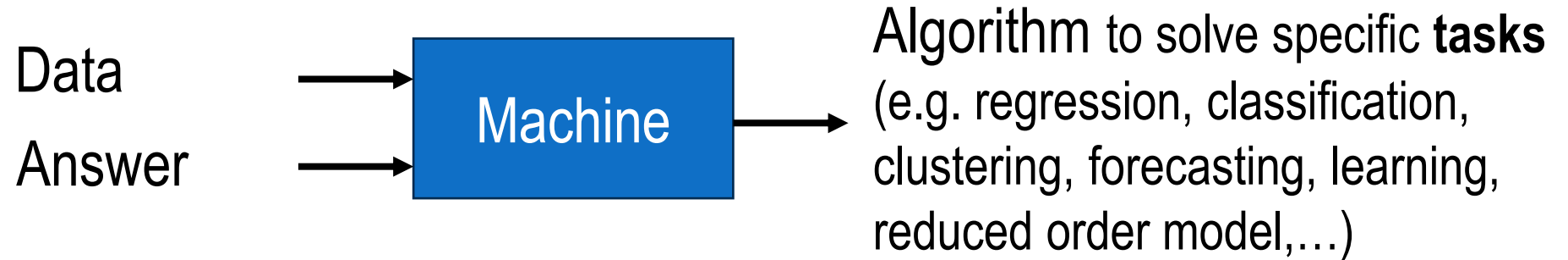
Physics-based vs Machine Learning (aka data-driven) models



Physics-based model analysis



Machine Learning



Building specialised models Investigating

- features (patterns in the data)
- correlations, relationships and dependencies
- associational evidence
- hidden structures

No causal relationships*



No (easy) distinction of confounding sources*

*Unless you use some advanced ML techniques

Leading poor generalization performance and physically inconsistent/improbable predictions

Definition of **Machine Learning** by Tom Mitchell (1998)

Study of algorithms that:

- Improve their **performance P**
- At some **task T**
- With **Experience E**



Selection of an appropriate performance metric (e.g. accuracy, squared error, cost/utility,...)



Function regression, classification, time-series forecasting, optimization...



Large amount of informative data

A well-defined learning task requires: P, T and E

Output: a data-driven model!

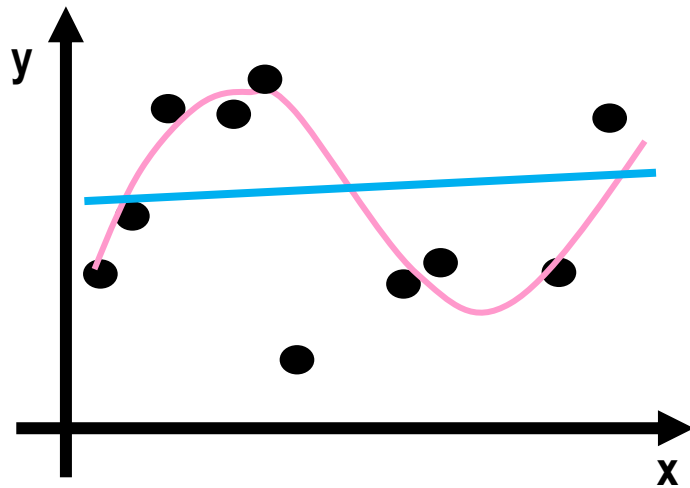
Traditional machine learning often gives a **single prediction**: one answer, one label, one number.

Deterministic (regression task): Data-driven models **chosen** and parameters **identified** to discover x-y relationships

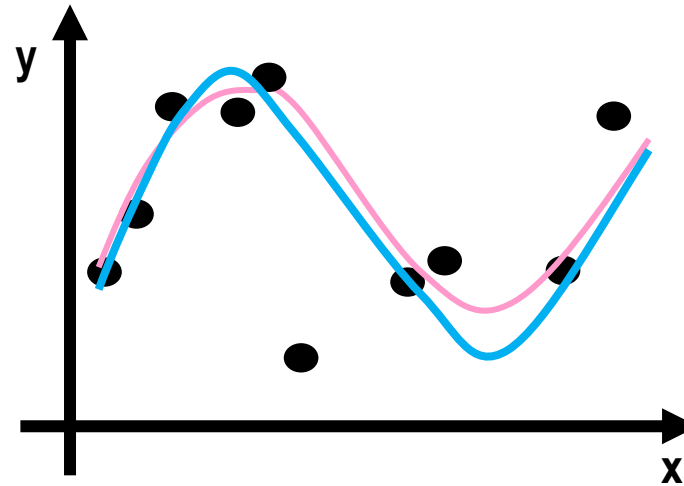
Unknown - Underlying correct x-y relationship

Approximated x-y relationship based on data

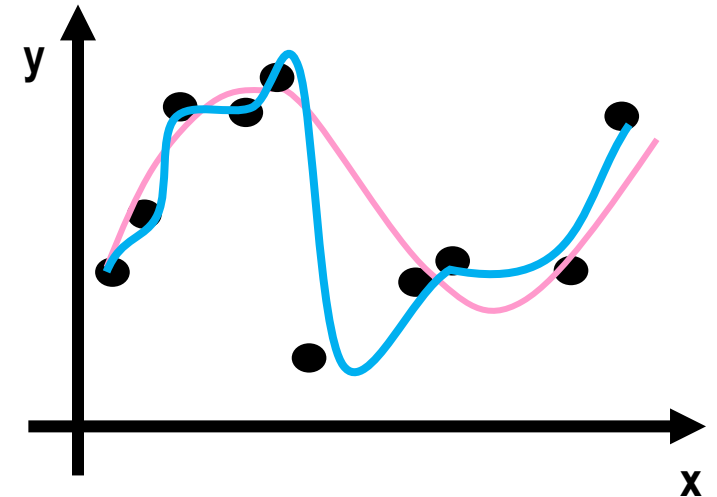
● Observed noisy and sparse data y



underfitting



good fitting



overfitting

- Models are obtained by minimizing an error on “**training-data**” → explaining the data
- Models' performance is assessed on unseen “**test-data**” → generalization
- **The loss/error will never be zero** - only for very trivial machine learning problems!
- **Occam's razor (aka principle of parsimony)**: less complex (simpler models) that give the same **prediction** (e.g., good fitting!), given the same data, are expected to generalize better

Why Probabilistic Machine Learning (PML)?

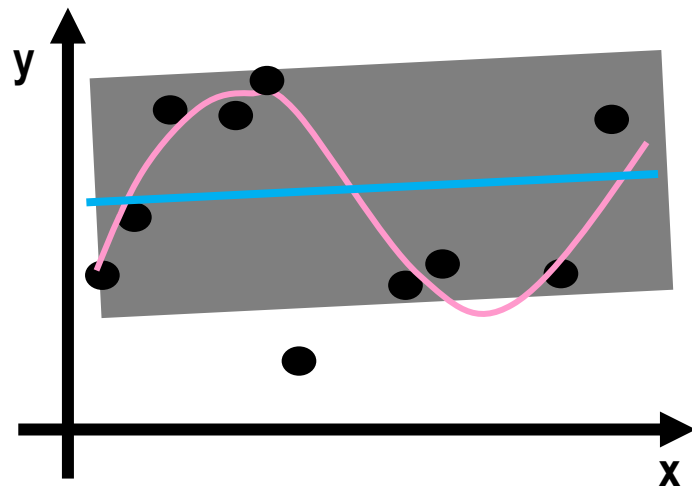
- **Critical tools when dealing with scarce, sparse and noisy data:**
 - Mitigates sensitivity to training data, where different datasets would lead to very different models
 - By embedding prior knowledge, they can help prevent wrong decisions based on unfounded extrapolations from insufficient data
- **Uncertainty Quantification:**
 - Model parameters and model output as a random variable described by a probability density function **instead of a point estimate**
 - Quantification when the model is guessing, affected by inherent input-output variability, or lacks data*
- **Guiding critical decision-making under uncertainty:**
 - Providing an estimate of confidence or uncertainty in predictions
 - Avoid results that are overconfident on out-of-distribution inputs
 - Overfitting: naturally penalised via priors
- **From the computational POV:**
 - **Less compute-intensive** to train and deploy
 - Less parameters to be optimized → More interpretable and transparent

*this will be come clearer when uncertainty sources will be described

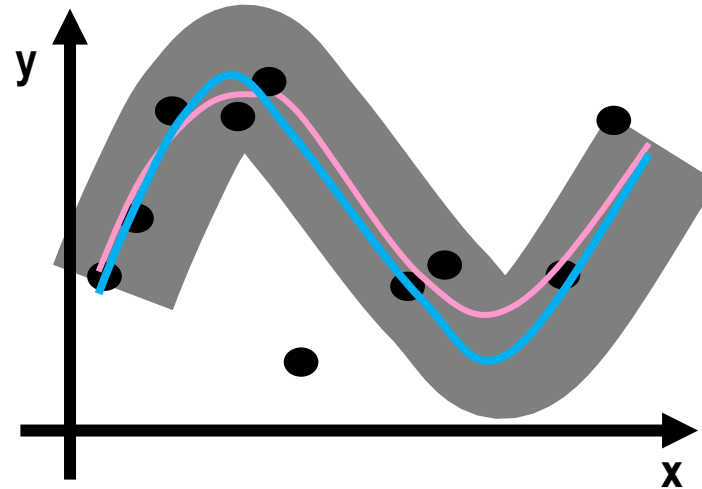
Probabilistic Machine Learning: quantifying uncertainty through probability distributions

Unknown - Underlying correct x-y relationship

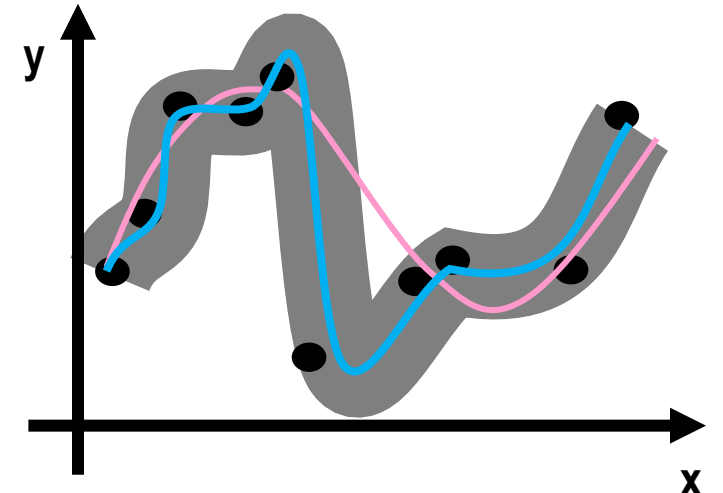
Approximated x-y relationship based on data



underfitting



good fitting



overfitting

● Observed noisy and sparse data y

Note: **Confidence intervals** (Frequentist) vs **Credible Intervals** (Bayesian) are not the same!

Frequentist: parameter of interest (e.g. mean) is considered fixed but unknown. The randomness comes solely from the data - no prior info is incorporated.

- Example: a 95% confidence interval indicates that in 95 out of 100 repetitions, the true parameter would be captured within the intervals calculated.
- Confidence intervals are not a PROBABILITY STATEMENT

Bayesian: probability statement about the parameter falling within the interval, given the data and prior information

- Example: a 95% Bayesian credible interval means there's a 95% probability that the true parameter value lies within that range.

Sources of uncertainties

Aleatoric uncertainty

Inherent variability (part-to-part, spatial, temporal, sensor-noise/measurement uncertainty, inputs, EOV)
(**irreducible uncertainty**)

Epistemic uncertainty

Lack of knowledge about a fixed model **parameter value** or about the **model itself** –the uncertainty can be reduced by carrying out some experiments/investigations
(**reducible uncertainty**)

- Ambient excitation, material scatter, turbulence, transduction noise.
- It cannot be reduced by collecting more data or by improving the model, although it can, in principle, be correctly quantified.

- lack of knowledge: limited or unrepresentative data, incomplete or idealised models, and unknown-but-fixed model parameters.
- It is, by definition, reducible — with perfect knowledge of the model and its parameters the epistemic component would vanish.

Typically, they are coupled during model building → **if the model form is wrong, the aleatoric uncertainty cannot be estimated correctly:** unmodelled physics leaks into the residual and inflates or biases what is reported as “noise”.

Aleatoric uncertainty

Inherent variability (part-to-part, spatial, temporal, sensor-noise/measurement uncertainty, inputs, EOV)

(**irreducible uncertainty**)

Epistemic uncertainty

Lack of knowledge about a fixed model **parameter value** or about the **model itself** –the uncertainty can be reduced by carrying out some experiments/investigations
(**reducible uncertainty**)

Irreducible model parameters

Lack of knowledge about a fixed model parameter value

no experiments/investigations can be carried out (e.g. destructive testing on bridges) to reduce uncertainty
(**irreducible uncertainty**)

Mapping uncertainties – decision support at design stage

Model discrepancy due to physics (Model form)

Model prediction inconsistent with reality because of missing physics, choosing the wrong model class and/or model solver/code
(**reducible uncertainty**)

Model discrepancy due to data (Model form)

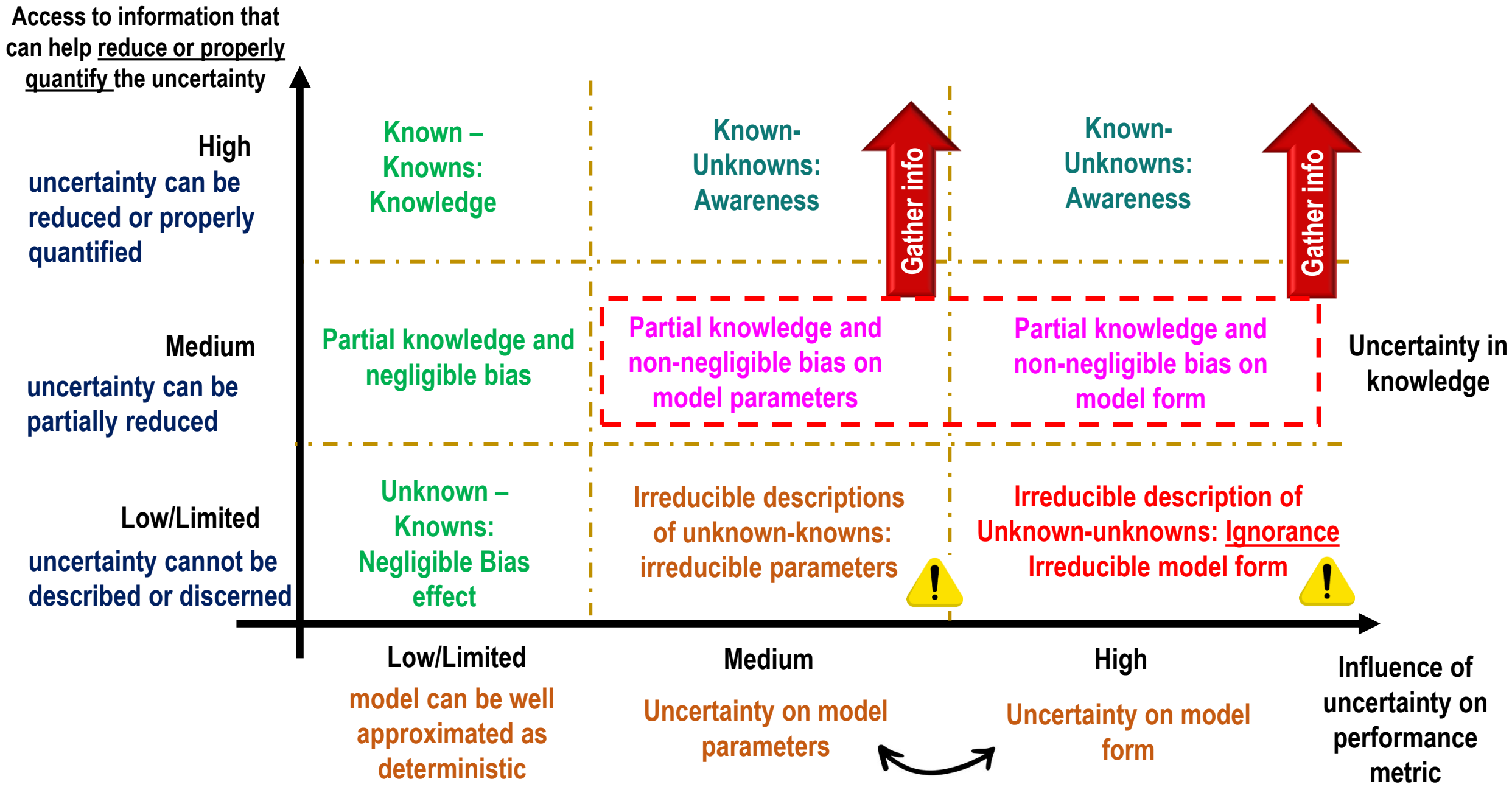
Model prediction inconsistent with reality because of corrupted/uninformative measurements, incorrect data-driven model
(**reducible uncertainty**)

Irreducible model form

Model prediction inconsistent with reality because of unknown unknowns (e.g. nonlinearity, discrepancy between training and testing data) and undistinguishable confounding sources
(**irreducible uncertainty**)

known unknowns

unknown unknowns: we don't know that we don't know!



Decision support in **operating conditions** - dealing with dynamic data:

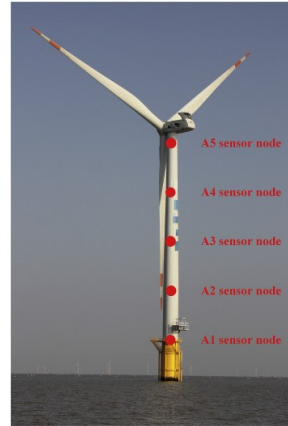
Digital twins – dynamic models integrating physics, data, and **uncertainty**

$\mathbf{X}_{true}$

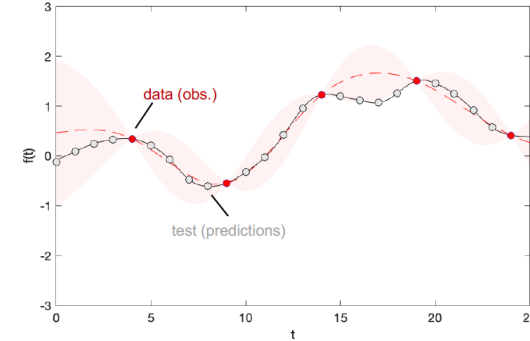
Input
(operational &
environmental
loading)



y_{true}

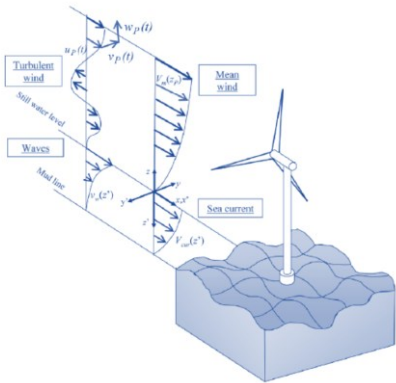


$y_{measured}$

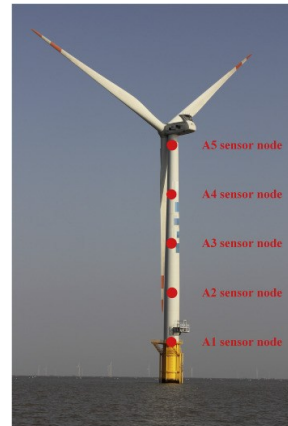


y_{data}

$\mathbf{X}_{phy}$



+



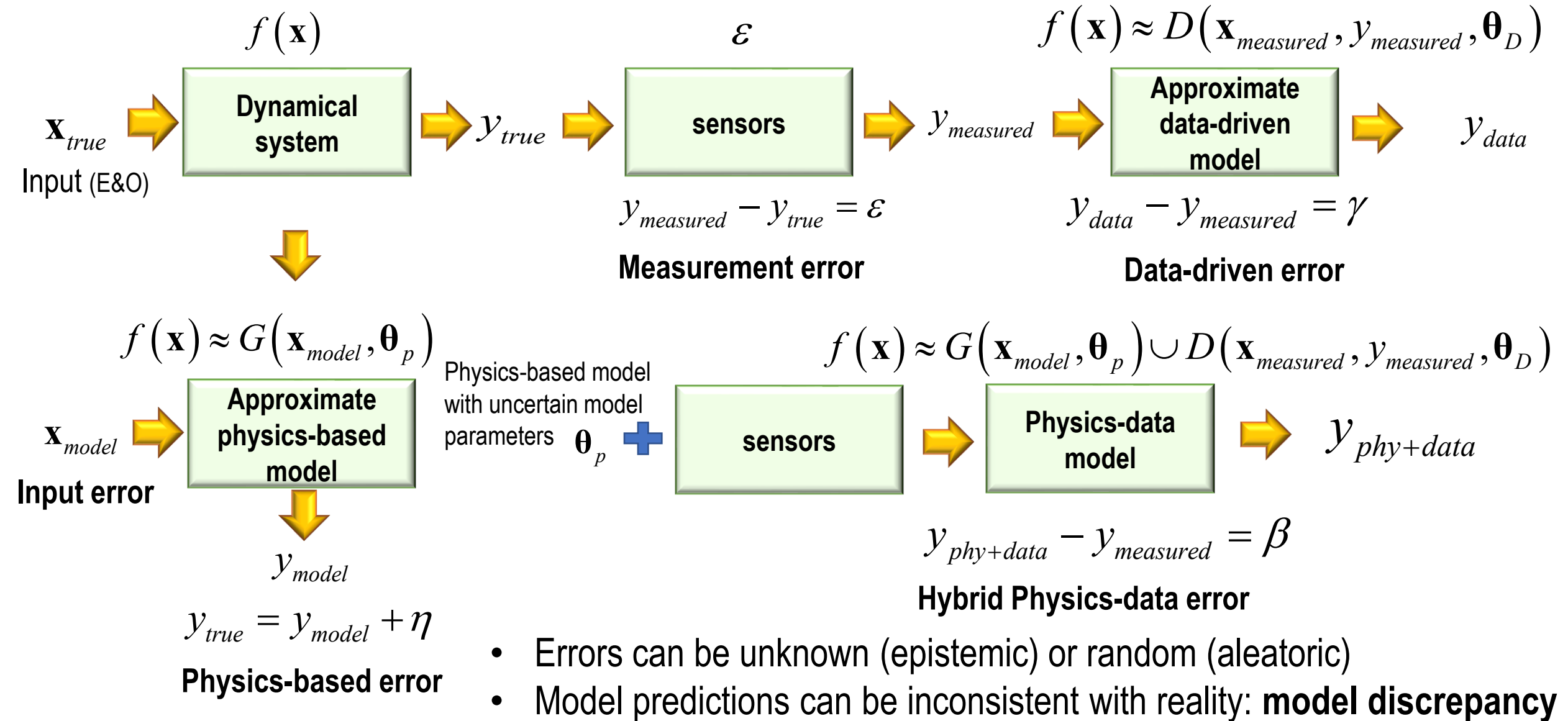
$y_{phy+data}$

The digital twin is a **time-evolving** digital model based on a **two-way interaction** between the physical system (usually monitored) and the digital counterpart, that can account for uncertainty, different systems interactions, and it is used for policy and decision making.

y_{phy}

$$y_{true} \neq y_{measured} \neq y_{learned} \neq y_{phy} \neq y_{data} \neq y_{phy+data}$$

What are (some of) the sources of uncertainty?



Uncertainties challenge from a **system** point of view

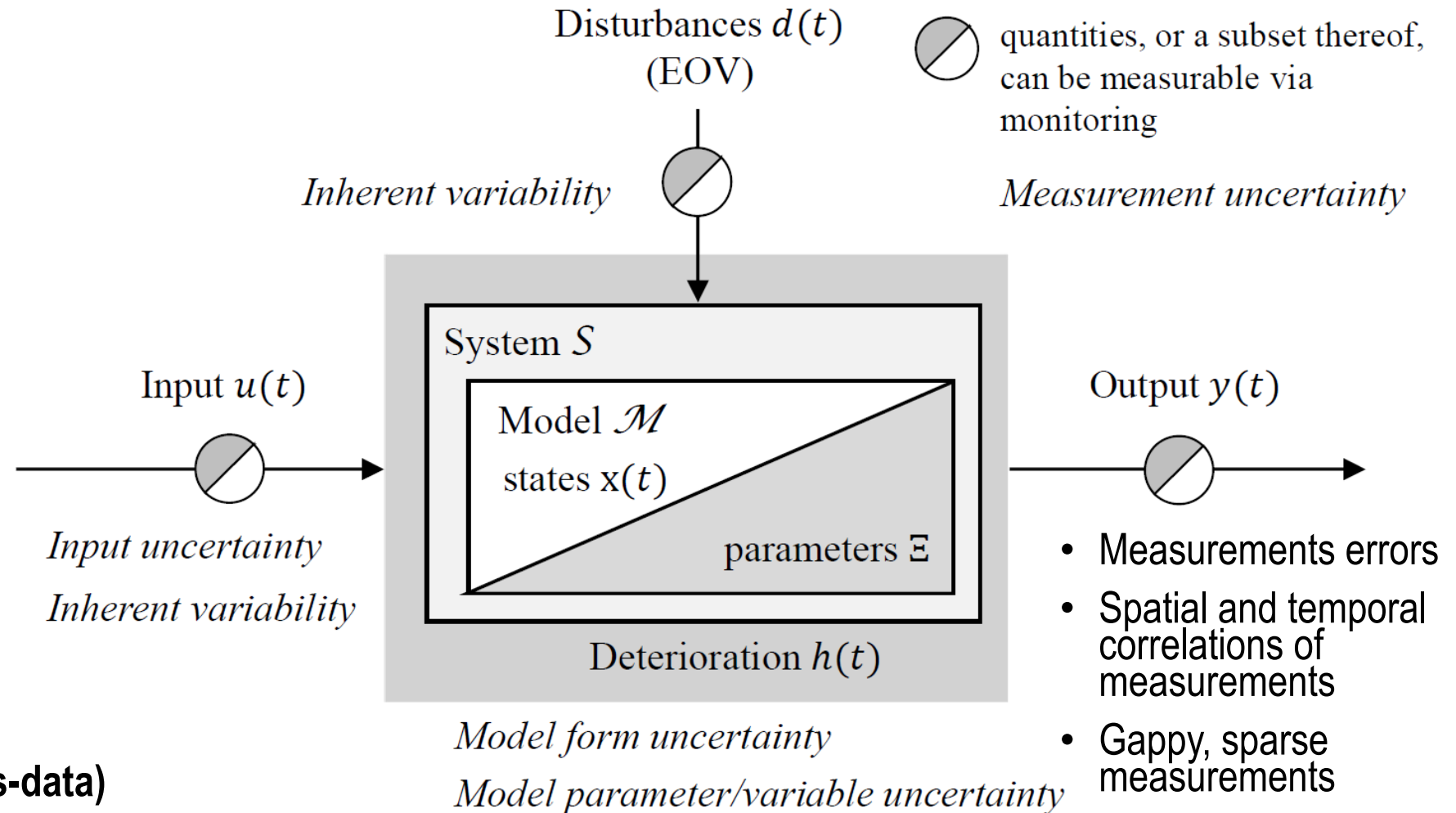
- **Uncertainty or inherent variability on:**

- **model parameters:**
geometry, material properties, boundary conditions
- **model inputs:**
environmental and operational loads

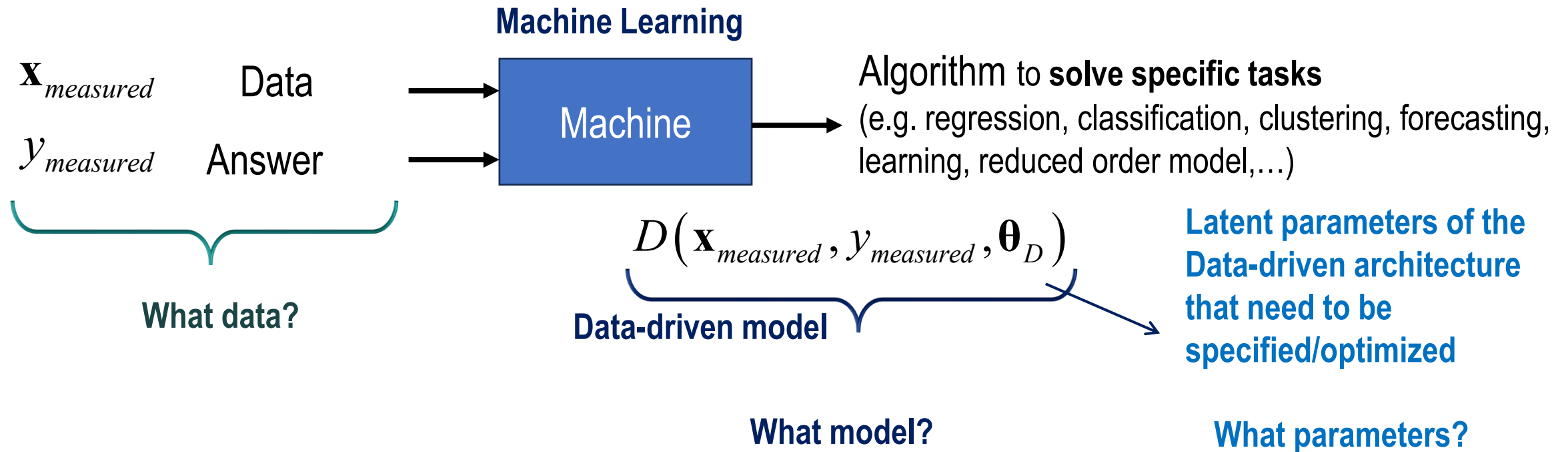
- Usually, a broad frequency range is involved → **Manufacturing variability** (Part-to-part variability, Spatial variability) becomes important

- **Model-form (e.g. hybrid physics-data)**

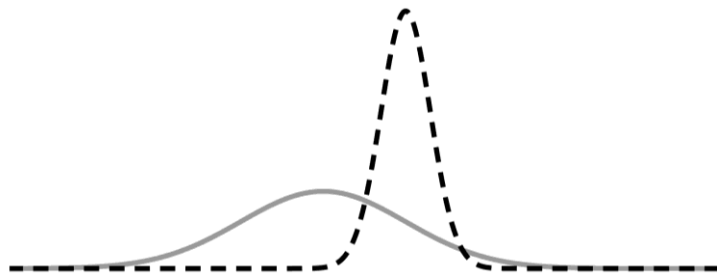
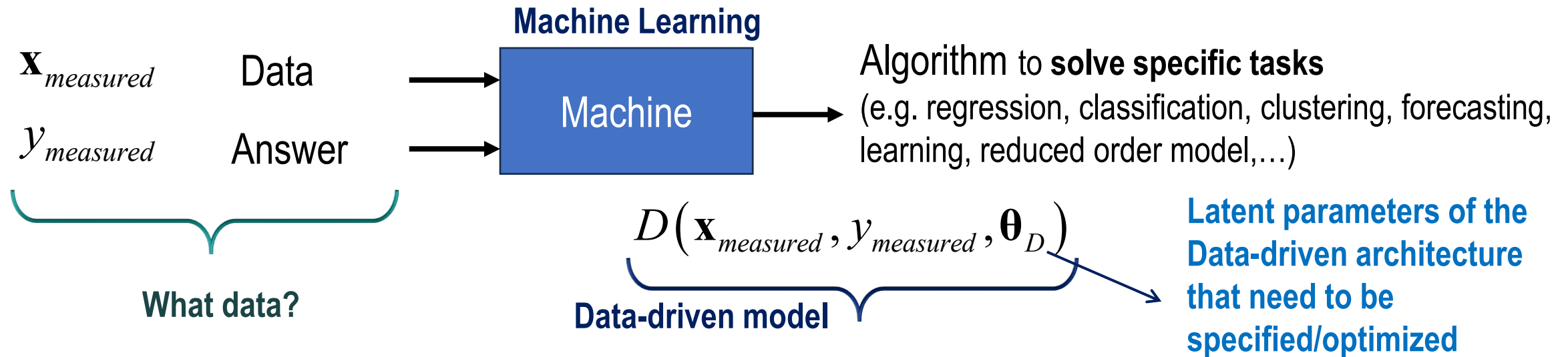
- **Domain knowledge**
- **solver/code**



Choices to be made in (deterministic) data-driven strategies



Uncertainties and challenges in (deterministic) data-driven architectures



Inherent data variability
(aleatory uncertainty)

Challenge 1: Do we
have enough
informative data?

Epistemic uncertainty - Model error due to:
Data:

- from aleatory uncertainty (not fully captured);
- From **corrupted** and/or non-informative data;

Model:

- from wrong model architecture
- and/or wrong model parameters

**Limiting the applicability of ML in guiding
decision making in Engineering**

Challenge 2: How do we
cope with uncertainty?

Challenge 3: Can the
model generalize to
unseen inputs?

Challenge 4: Can the
model be explainable &
interpretable?

Source of uncertainties matter, because they affect the meaning of the **prediction interval!**

- The same nominal 95% prediction interval **can represent entirely different things**,
- Therefore, it would justify **entirely different decisions**, depending on which uncertainty sources have been accounted for!
- An interval (of a specific width) might reflect:
 - only measurement noise (**aleatoric**) makes a confident statement about a well-characterised system;
 - only limited data (**epistemic**) is an admission of ignorance that further measurement could shrink;
 - an interval that omits **model-form uncertainty** may be narrow and wrong.
- Calibration metrics (coverage, negative log-likelihood, expected calibration error) assess whether reported uncertainty matches observed error frequencies, **but they do not by themselves reveal which source of uncertainty is being represented**

**Too many words..
Let's see this in
action!**

**L1_fileA_Prediction_Intervals
_Three_Meanings.ipynb**

For physics-based models: uncertainty in **model parameters**

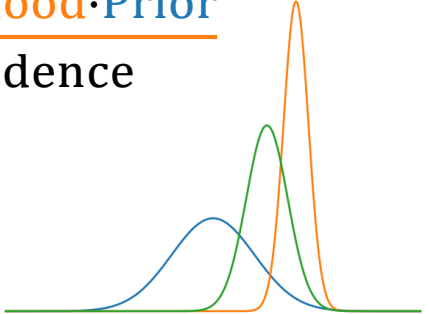
Can be handled by

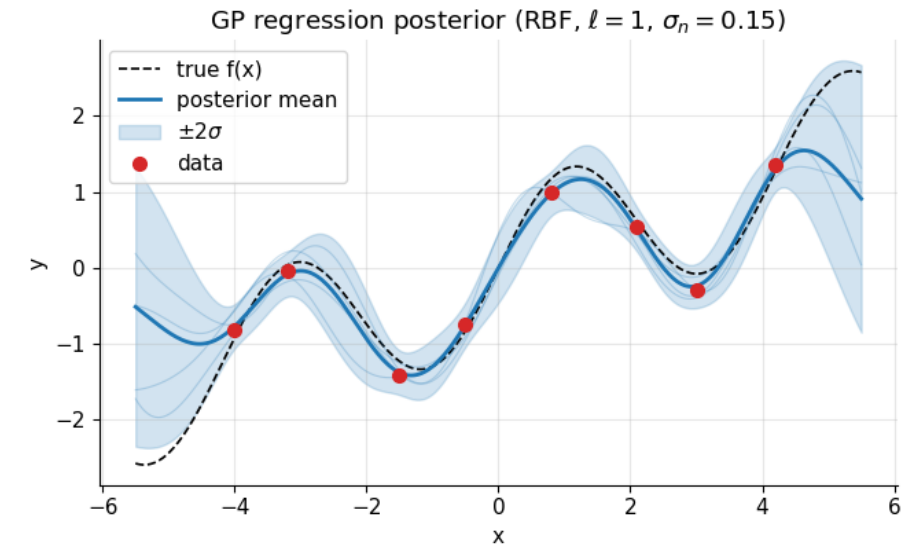
- **Bayesian inference**

- yields a posterior distribution over the **latent parameters** given the data, a physics-based forward model, an error (likelihood) model and a prior.
- by using sampling (e.g. Markov-chain Monte Carlo) or variational inference

• **Gaussian processes (GPs)**

- offer a complementary, non-parametric Bayesian route:
- a GP places a distribution over functions and returns a predictive mean with a variance that grows where data are sparse, so the variance encodes epistemic uncertainty about the response;
- used as an emulator or as an additive discrepancy term within Bayesian calibration,
- a GP also supports **inference about fixed-but-unknown physical parameters**

$$\text{Posterior} = \frac{\text{Likelihood} \cdot \text{Prior}}{\text{Evidence}}$$




Next Lecture

For data-driven models: Uncertainty in the trainable parameters (e.g. weights)

can be handled by

- **Bayesian neural networks**
 - treat the weights as random variables and infer their posterior,
- **Deep ensembles**
 - which train several independently initialised networks on resampled data;

Both yield a predictive distribution — a posterior or a mean with bounds — over model outputs, and, with **an appropriate output head**, can separate aleatoric from epistemic contributions

Fundamentals of Bayesian inference:

all done in the notebook:

Fundamental of Probabilistic Machine Learning: Introduction to Bayesian Inference

Instructor: Alice Cicirello

Generated for Training school scientific machine learning for digital twins

The first draft version of the notebook was generated using Anthropic's Claude

Date: September 2026

Last Modifications and checks made by Alice Cicirello on: 5/09/2026

Short description This notebook is a self-contained introduction to *Bayesian inference*: what it is, the statistics you need to understand it, how it differs from the *frequentist* approach you have probably seen in undergraduate courses, and how to actually *do* it on engineering problems.

Pre-requisites comfortable with calculus, basic linear algebra, and a little Python. No prior exposure to probability theory is required — we build it from the ground up.

In this lecture you learned about

- Machine Learning vs Probabilistic Machine Learning
- Sources of uncertainties (with notebook)
- Fundamental of probabilistic analysis (all in the notebook)

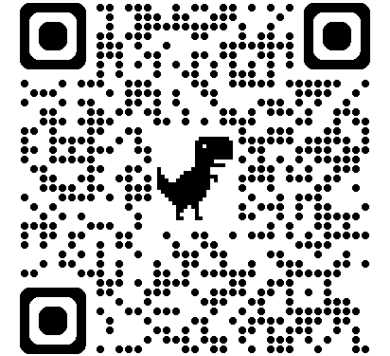
Next Lecture

(mostly) hands-on session on

- Bayesian Model updating
- Gaussian Process for Regression
- System Identification using probabilistic machine learning

Further readings

- **Probabilistic Machine Learning: An Introduction, K. P. Murphy**
(<https://probml.github.io/pml-book/book1.html>)
- **Deep Learning: foundations and concepts, C. Bishop, H. Bishop**
(<https://www.bishopbook.com/>)
- **Pattern Recognition and machine learning, C. Bishop**
(<https://www.microsoft.com/en-us/research/uploads/prod/2006/01/Bishop-Pattern-Recognition-and-Machine-Learning-2006.pdf>)
- **Information Theory, Inference, and Learning Algorithms, D. McKay**
(<https://www.inference.org.uk/mackay/itila/book.html>)



I attempted to credit all the content/figures where I could find the original source. Sometimes the material is so widely used that it's difficult to be sure of the original author. Please let me know if you find missing references.